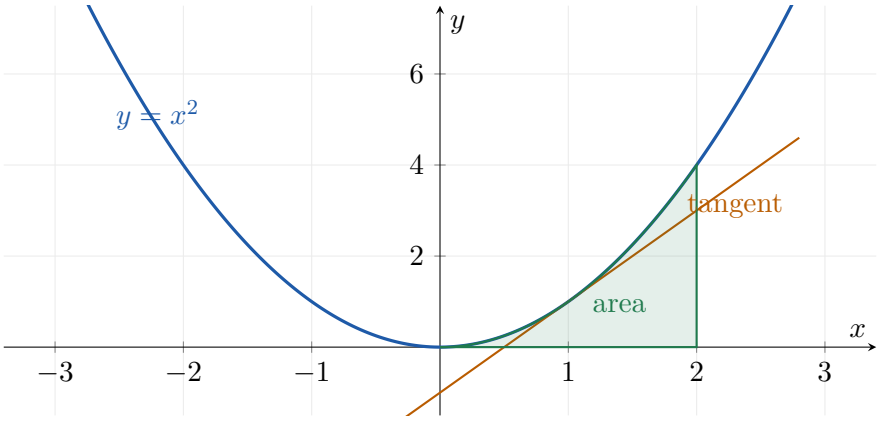
Differential and Integral Calculus for Absolute Beginners

Lecture notes for computer science students who need a slow start



May 16, 2026

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How to use these notes

These notes are intentionally slow. They do not assume that the reader already knows calculus. The goal is not to impress the reader with advanced notation. The goal is to build usable mathematical understanding from very small steps.

Big idea

Calculus has two main questions.

1. **Differential calculus:** how fast does something change at one point?
2. **Integral calculus:** how much has accumulated over an interval?

The derivative answers the first question. The integral answers the second question.

In computer science, calculus appears in optimization, machine learning, graphics, numerical methods, simulations, data modelling, probability, and algorithms. You do not need to love calculus to use it. But you need to understand what the symbols mean.

What this course tries to avoid

- memorizing formulas without knowing what they do;
- using notation before explaining it;
- jumping from a definition directly to a difficult problem;
- pretending that algebra mistakes are "small" mistakes. In calculus, algebra mistakes usually destroy the answer.

1 Before calculus: functions, graphs, and notation

1.1 What is a function?

A function is a rule that takes an input and produces exactly one output.

Function

A function f from a set of inputs to a set of outputs is a rule such that every allowed input x has exactly one output $f(x)$.

Examples:

$$f(x) = x^2, \qquad g(x) = 2x + 1, \qquad h(x) = \sqrt{x}.$$

If $f(x) = x^2$, then:

$$f(3) = 3^2 = 9, \qquad f(-4) = (-4)^2 = 16.$$

Common mistake

The expression $f(x)$ does not mean $f \cdot x$. It means: the value of the function f at the input x.

1.2 Input, output, domain, range

The **domain** is the set of inputs for which the function makes sense. The **range** is the set of outputs that the function can produce.

For example, for

$$h(x) = \sqrt{x},$$

we cannot use negative real inputs if we work only with real numbers. Therefore the domain is

$$[0, \infty).$$

Example: domain

Find the domain of

$$f(x) = \frac{1}{x-2}.$$

The denominator cannot be zero, so

$$x - 2 \neq 0 \quad \Rightarrow \quad x \neq 2.$$

Therefore the domain is

$$\mathbb{R} \setminus \{2\}.$$

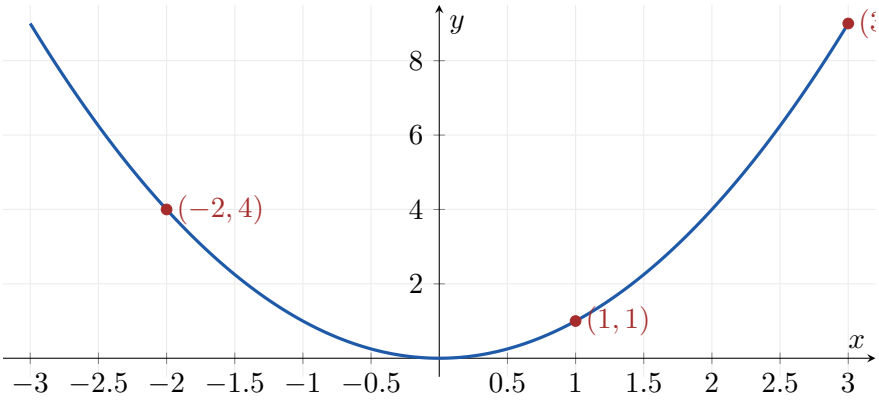
1.3 Graphs

A graph of a function is a picture of all points

$$(x, f(x)).$$

For instance, the graph of $f(x) = x^2$ consists of all points

$$(x, x^2).$$



1.4 Slope of a straight line

For a straight line, slope measures how much y changes when x changes.

Slope of a line

If a line passes through points (x_1, y_1) and (x_2, y_2), its slope is

$$m = \frac{y_2 - y_1}{x_2 - x_1}.$$

The numerator is the change in y. The denominator is the change in x.

Example: slope

Find the slope of the line through $(1, 2)$ and $(4, 8)$.

$$m = \frac{8-2}{4-1} = \frac{6}{3} = 2.$$

This means: when x increases by 1, y increases by 2.

Common mistake

Do not swap the order in only one part of the fraction. Both differences must use the same order:

$$\frac{y_2 - y_1}{x_2 - x_1} \quad \text{or} \quad \frac{y_1 - y_2}{x_1 - x_2}.$$

But not

$$\frac{y_2 - y_1}{x_1 - x_2}.$$

Practice

Find the slope of the line through each pair of points.

1. $(0, 1)$ and $(2, 5)$
2. $(-1, 3)$ and $(3, -5)$
3. $(2, 7)$ and $(2, 10)$

What happens in the third case?

2 Limits: the language of approaching

2.1 The informal idea

A limit describes what value a function approaches when the input approaches a given number.

Informal meaning of limit

The notation

$$\lim_{x \to a} f(x) = L$$

means: when x gets closer and closer to a, the values $f(x)$ get closer and closer to L.

The function does not necessarily have to be defined at $x = a$. The limit talks about what happens **near** a, not necessarily exactly at a.

2.2 A first example

Consider

$$f(x) = x + 2.$$

As $x \to 3$, the output approaches

$$3 + 2 = 5.$$

Therefore

$$\lim_{x \to 3} (x + 2) = 5.$$

This example is easy because the function is continuous. Later we will give that word a precise meaning.

2.3 A limit from a table

Let

$$f(x) = \frac{x^2 - 1}{x - 1}.$$

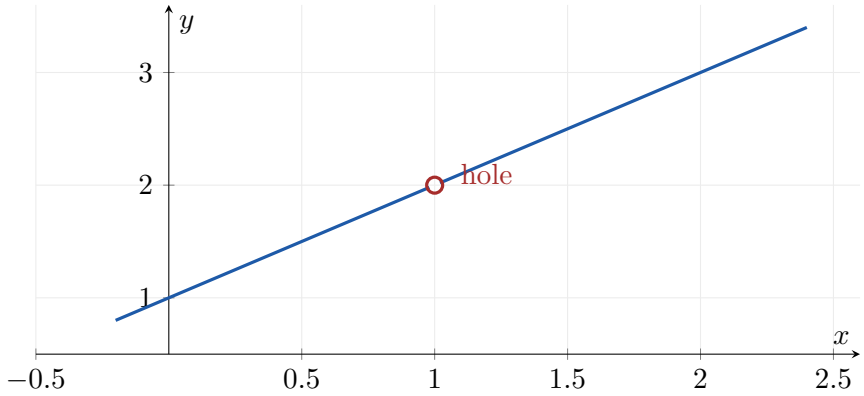
At $x = 1$, the expression gives $0/0$, so it is not defined. But for $x \neq 1$, we can simplify:

$$\frac{x^2 - 1}{x - 1} = \frac{(x - 1)(x + 1)}{x - 1} = x + 1.$$

Near $x = 1$, the function behaves like $x + 1$, so the limit is

$$\lim_{x \to 1} \frac{x^2 - 1}{x - 1} = 2.$$

x	0.9	0.99	0.999	1.001	1.01	1.1
$f(x)$	1.9	1.99	1.999	2.001	2.01	2.1



Common mistake

The expression $0/0$ is not equal to zero. It is undefined. In limits, $0/0$ usually means: simplify the expression before deciding what happens.

2.4 Left-hand and right-hand limits

Sometimes the behavior from the left and from the right is different.

One-sided limits

$$\lim_{x \to a^-} f(x)$$

means that x approaches a from values smaller than a.

$$\lim_{x \to a^+} f(x)$$

means that x approaches a from values greater than a.

A two-sided limit exists only if both one-sided limits exist and are equal.

Jump example

Let

$$f(x) = \begin{cases} 1, & x < 0, \\ 3, & x \geq 0. \end{cases}$$

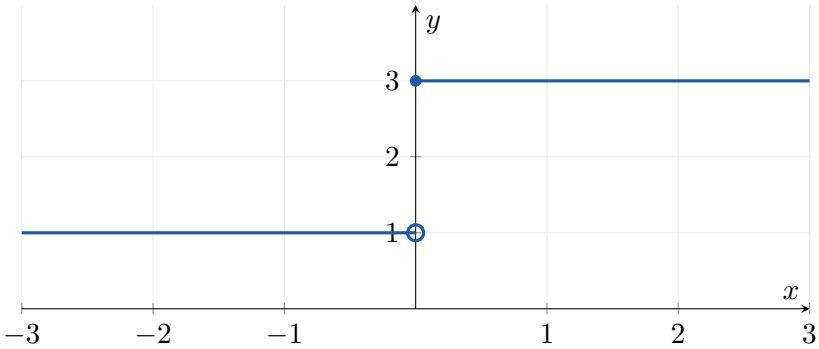
Then

$$\lim_{x \to 0^-} f(x) = 1, \qquad \lim_{x \to 0^+} f(x) = 3.$$

Since the two values are different,

$$\lim_{x \to 0} f(x)$$

does not exist.



2.5 Basic limit rules

If the limits exist, then:

$$\lim_{x\to a}(f(x)+g(x)) = \lim_{x\to a} f(x) + \lim_{x\to a} g(x),$$

$$\lim_{x\to a}(cf(x)) = c\lim_{x\to a} f(x),$$

$$\lim_{x\to a}(f(x)g(x)) = \left(\lim_{x\to a} f(x)\right)\left(\lim_{x\to a} g(x)\right),$$

$$\lim_{x\to a}\frac{f(x)}{g(x)} = \frac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)} \quad \text{if } \lim_{x\to a} g(x) \neq 0.$$

2.6 Limits of polynomials and rational functions

For a polynomial, limits are simple: substitute the value.

Polynomial limit

$$\lim_{x\to 2}(x^3 - 4x + 1) = 2^3 - 4\cdot 2 + 1 = 8 - 8 + 1 = 1.$$

For a rational function, first check whether the denominator becomes zero.

Rational limit without problem

$$\lim_{x\to 1}\frac{x+5}{x^2+1} = \frac{1+5}{1^2+1} = \frac{6}{2} = 3.$$

Rational limit with cancellation

$$\lim_{x\to 2}\frac{x^2-4}{x-2}.$$

Direct substitution gives 0/0. Factor:

$$\frac{x^2-4}{x-2} = \frac{(x-2)(x+2)}{x-2} = x+2 \quad (x \neq 2).$$

Therefore

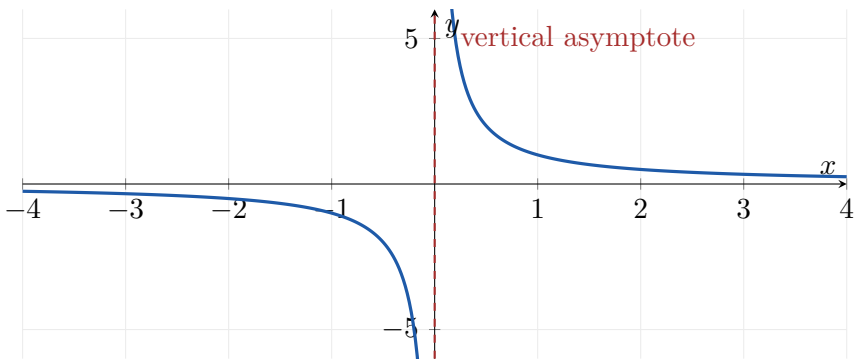
$$\lim_{x\to 2}\frac{x^2-4}{x-2} = 4.$$

2.7 Infinite limits and vertical asymptotes

If function values grow without bound as x approaches a, we write an infinite limit.

Example:

$$\lim_{x\to 0^+} \frac{1}{x} = +\infty, \qquad \lim_{x\to 0^-} \frac{1}{x} = -\infty.$$



Common mistake

The symbol ∞ is not a real number. Writing $f(x) \to \infty$ means that the function grows without any finite bound.

2.8 Limits at infinity and horizontal asymptotes

The notation

$$\lim_{x\to\infty} f(x) = L$$

means that as x becomes very large, the function values approach L.

For example:

$$\lim_{x\to\infty} \frac{1}{x} = 0.$$

The line $y = 0$ is a horizontal asymptote.

Practice

Compute the following limits.

1. $\lim_{x\to 4}(3x - 7)$
2. $\lim_{x\to -1}(x^2 + 2x + 5)$
3. $\lim_{x\to 3} \frac{x^2 - 9}{x - 3}$
4. $\lim_{x\to 0^+} \frac{1}{x^2}$
5. $\lim_{x\to\infty} \frac{5}{x}$

3 Continuity: functions without breaks

3.1 The intuitive idea

A function is continuous if its graph can be drawn without lifting the pencil. This is a useful picture, but the mathematical definition is more precise.

Continuity at a point

A function f is continuous at $x = a$ if all three conditions hold:

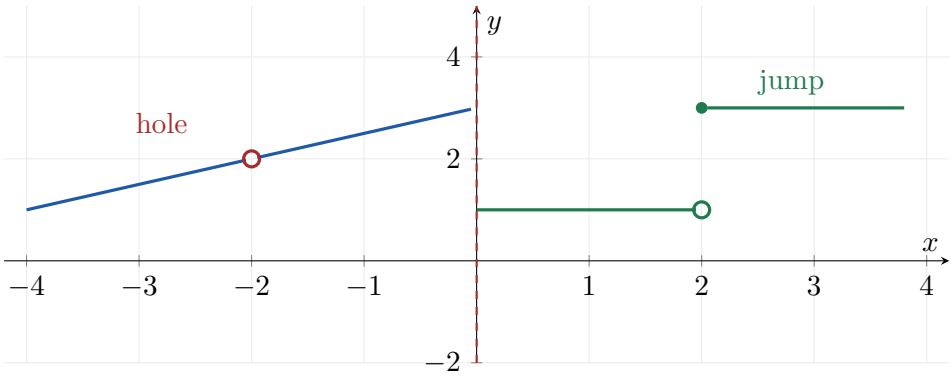
1. $f(a)$ is defined;
2. $\lim_{x \to a} f(x)$ exists;
3. $\lim_{x \to a} f(x) = f(a)$.

So continuity means: the value at the point agrees with the value approached near the point.

3.2 Three ways continuity can fail

Continuity can fail if:

1. the function has a hole;
2. the function has a jump;
3. the function blows up to infinity.



3.3 Continuous functions you should recognize

The following functions are continuous on their natural domains:

- polynomials, such as $x^2 + 3x - 1$;
- rational functions, where the denominator is not zero;
- square root functions, where the expression under the root is nonnegative;
- trigonometric functions $\sin x$, $\cos x$;
- exponential functions a^x, where $a > 0$;
- logarithmic functions $\ln x$, where $x > 0$.

Continuity check

Is

$$f(x) = \frac{x+1}{x-4}$$

continuous at $x = 2$?
Yes. The denominator is not zero at $x = 2$, because $2 - 4 = -2$. Therefore the rational function is continuous at $x = 2$.

Not continuous

Is the same function continuous at $x = 4$?
No. The denominator is zero at $x = 4$, so $f(4)$ is not defined.

3.4 The Intermediate Value Theorem

Intermediate Value Theorem

If f is continuous on $[a, b]$, and N is between $f(a)$ and $f(b)$, then there exists at least one $c \in [a, b]$ such that

$$f(c) = N.$$

In plain words: a continuous function cannot jump over a value.

Root existence

Let

$$f(x) = x^3 - x - 1.$$

Then

$$f(1) = 1 - 1 - 1 = -1, \qquad f(2) = 8 - 2 - 1 = 5.$$

The function is continuous, and it changes sign between 1 and 2. Therefore there is at least one root in $(1, 2)$.

This idea is the basis of the bisection method in numerical computing.

Practice

For each function, decide where it is continuous.

1. $f(x) = x^2 - 5x + 6$
2. $g(x) = \dfrac{1}{x+3}$
3. $h(x) = \sqrt{x-2}$
4. $p(x) = \dfrac{x^2 - 1}{x - 1}$

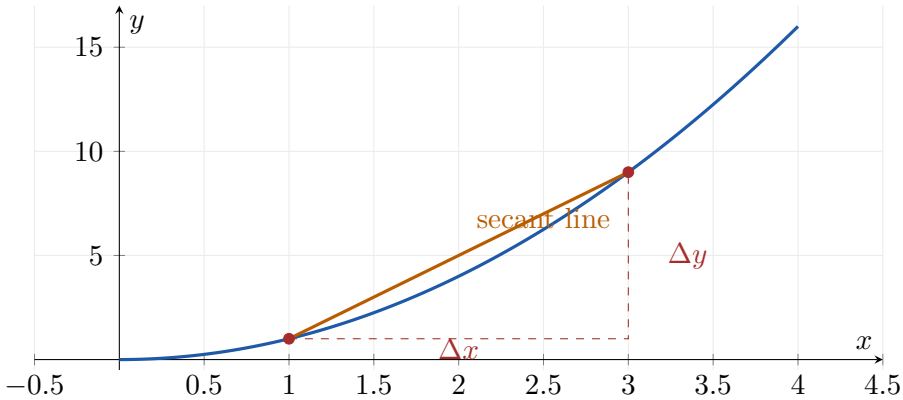
4 Derivatives: measuring change at one point

4.1 Average rate of change

Before the derivative, we need the average rate of change. For a function f, the average rate of change from $x = a$ to $x = b$ is

$$\frac{f(b) - f(a)}{b - a}.$$

This is the slope of the secant line through the two graph points.



For $f(x) = x^2$, from $x = 1$ to $x = 3$:

$$\frac{f(3) - f(1)}{3 - 1} = \frac{9 - 1}{2} = 4.$$

4.2 Instantaneous rate of change

The derivative is the rate of change at one point. To get it, we take average rates of change over shorter and shorter intervals.

Derivative at a point

The derivative of f at $x = a$ is

$$f'(a) = \lim_{h \to 0} \frac{f(a + h) - f(a)}{h},$$

if this limit exists.

The fraction

$$\frac{f(a + h) - f(a)}{h}$$

is called the difference quotient.

4.3 Computing a derivative from the definition

Derivative of x^2 from the definition

Let $f(x) = x^2$. Compute $f'(a)$.
Start with the definition:

$$f'(a) = \lim_{h\to 0} \frac{f(a+h) - f(a)}{h}.$$

Compute each part:

$$f(a+h) = (a+h)^2 = a^2 + 2ah + h^2, \qquad f(a) = a^2.$$

Subtract:

$$f(a+h) - f(a) = a^2 + 2ah + h^2 - a^2 = 2ah + h^2.$$

Divide by h:

$$\frac{2ah + h^2}{h} = 2a + h.$$

Take the limit:

$$f'(a) = \lim_{h\to 0}(2a + h) = 2a.$$

Therefore

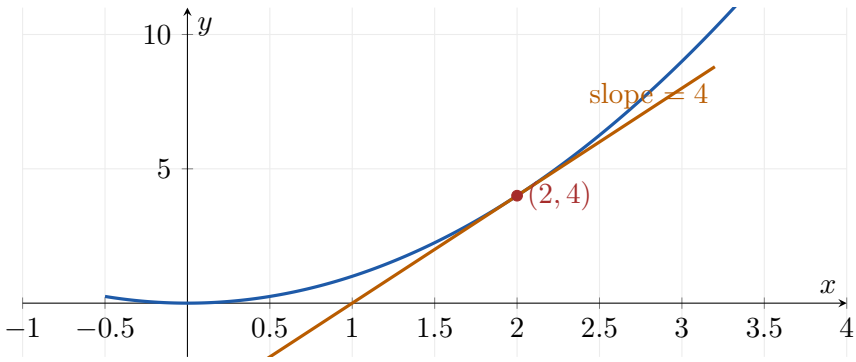
$$\boxed{\frac{\mathrm{d}}{\mathrm{d}x}x^2 = 2x.}$$

Common mistake

The variable h is not zero during the simplification. We divide by h before taking the limit $h \to 0$. Do not substitute $h = 0$ too early.

4.4 Derivative as slope of the tangent line

The derivative $f'(a)$ is the slope of the tangent line to the graph at $x = a$.



For $f(x) = x^2$, we have $f'(x) = 2x$. At $x = 2$,

$$f'(2) = 4.$$

So the tangent line has slope 4.

4.5 Derivative notation

The derivative of $y = f(x)$ can be written as:

$$f'(x), \qquad y', \qquad \frac{\mathrm{d}y}{\mathrm{d}x}, \qquad \frac{\mathrm{d}}{\mathrm{d}x} f(x).$$

All of these describe the same idea: how y changes with respect to x.

Common mistake

The notation $\frac{\mathrm{d}y}{\mathrm{d}x}$ is not an ordinary fraction, but it behaves like a ratio in many useful ways. It means "the derivative of y with respect to x".

Practice

Use the definition to compute the derivative.

1. $f(x) = 3x + 1$
2. $f(x) = x^2 + 1$
3. $f(x) = x^3$ hint: $(a + h)^3 = a^3 + 3a^2h + 3ah^2 + h^3$

5 Basic derivative rules

Once we know what a derivative means, we need rules for computing it efficiently.

5.1 Constant rule

If c is a constant, then

$$\frac{\mathrm{d}}{\mathrm{d}x}c = 0.$$

A constant function does not change, so its rate of change is zero.

5.2 Power rule

Power rule

For real powers where the expression is defined,

$$\frac{\mathrm{d}}{\mathrm{d}x}x^n = nx^{n-1}.$$

Examples:

$$\frac{\mathrm{d}}{\mathrm{d}x}x^5 = 5x^4, \qquad \frac{\mathrm{d}}{\mathrm{d}x}x^2 = 2x, \qquad \frac{\mathrm{d}}{\mathrm{d}x}x = 1.$$

Negative powers

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{x}\right) = \frac{\mathrm{d}}{\mathrm{d}x}x^{-1} = -x^{-2} = -\frac{1}{x^2}.$$

Fractional powers

$$\frac{\mathrm{d}}{\mathrm{d}x}\sqrt{x} = \frac{\mathrm{d}}{\mathrm{d}x}x^{1/2} = \frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}.$$

5.3 Constant multiple rule

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(cf(x)\right) = cf'(x).$$

Example:

$$\frac{\mathrm{d}}{\mathrm{d}x}(7x^3) = 7 \cdot 3x^2 = 21x^2.$$

5.4 Sum and difference rules

$$\frac{\mathrm{d}}{\mathrm{d}x}(f(x) + g(x)) = f'(x) + g'(x),$$
$$\frac{\mathrm{d}}{\mathrm{d}x}(f(x) - g(x)) = f'(x) - g'(x).$$

Polynomial derivative

Find the derivative of

$$f(x) = 4x^5 - 3x^2 + 7x - 11.$$

Differentiate term by term:

$$f'(x) = 20x^4 - 6x + 7.$$

5.5 Product rule

Product rule

If $h(x) = f(x)g(x)$, then

$$h'(x) = f'(x)g(x) + f(x)g'(x).$$

Product rule

Let

$$h(x) = x^2 \sin x.$$

Then

$$h'(x) = 2x \sin x + x^2 \cos x.$$

Common mistake

The derivative of a product is not usually the product of derivatives:

$$(fg)' \neq f'g'.$$

Use the product rule.

5.6 Quotient rule

Quotient rule

If

$$h(x) = \frac{f(x)}{g(x)},$$

then

$$h'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2},$$

where $g(x) \neq 0$.

Quotient rule

Let

$$h(x) = \frac{x^2 + 1}{x - 1}.$$

Then

$$h'(x) = \frac{2x(x-1) - (x^2+1) \cdot 1}{(x-1)^2}.$$

Simplify:

$$h'(x) = \frac{2x^2 - 2x - x^2 - 1}{(x-1)^2} = \frac{x^2 - 2x - 1}{(x-1)^2}.$$

5.7 Chain rule

The chain rule is used when one function is inside another function.

Chain rule

If

$$y = f(g(x)),$$

then

$$\frac{\mathrm{d}y}{\mathrm{d}x} = f'(g(x))g'(x).$$

In words: derivative of the outside, evaluated at the inside, times derivative of the inside.

Chain rule with a power

Find the derivative of

$$y = (3x+1)^5.$$

The outside function is u^5. The inside function is $u = 3x+1$.

$$y' = 5(3x+1)^4 \cdot 3 = 15(3x+1)^4.$$

Chain rule with sine

$$\frac{\mathrm{d}}{\mathrm{d}x}\sin(x^2) = \cos(x^2)\cdot 2x.$$

Common mistake

The most common chain rule mistake is forgetting to multiply by the derivative of the inside function.

$$\frac{\mathrm{d}}{\mathrm{d}x}(3x+1)^5 \neq 5(3x+1)^4.$$

The correct answer is

$$15(3x+1)^4.$$

5.8 Derivatives of common functions

Function	Derivative
x^n	nx^{n-1}
e^x	e^x
a^x	$a^x \ln a$
$\ln x$	$1/x$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$

Practice

Compute the derivatives.

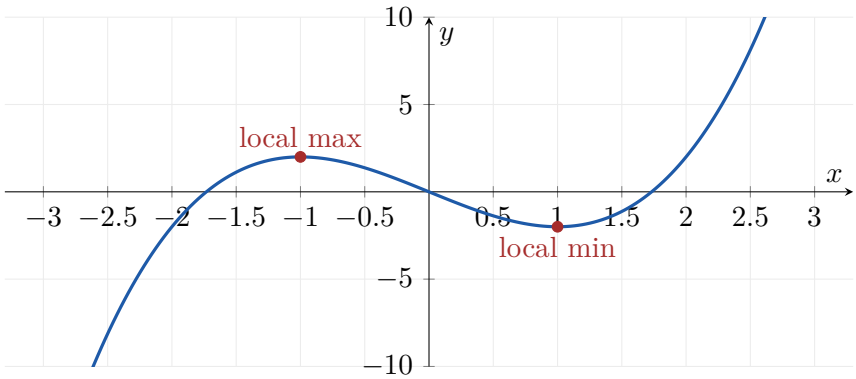
1. $f(x) = 5x^4 - 2x + 9$
2. $g(x) = (x^2+1)^3$
3. $h(x) = x^2 e^x$

4. $p(x) = \dfrac{\ln x}{x}$
5. $q(x) = \cos(4x)$

6 What derivatives tell us about graphs

6.1 Increasing and decreasing functions

If $f'(x) > 0$, then the function is increasing. If $f'(x) < 0$, then the function is decreasing.



For

$$f(x) = x^3 - 3x,$$

we get

$$f'(x) = 3x^2 - 3 = 3(x-1)(x+1).$$

The derivative is zero at $x = -1$ and $x = 1$. These are candidates for local extrema.

6.2 Critical points

Critical point

A critical point is a point where

$$f'(x) = 0$$

or where $f'(x)$ does not exist.

Critical points are important because local maxima and minima can occur there.

6.3 First derivative test

To classify a critical point, look at the sign of $f'(x)$ on the left and right.

Before point	At point	After point	Meaning
$+$	0	$-$	local maximum
$-$	0	$+$	local minimum
$+$	0	$+$	no max/min
$-$	0	$-$	no max/min

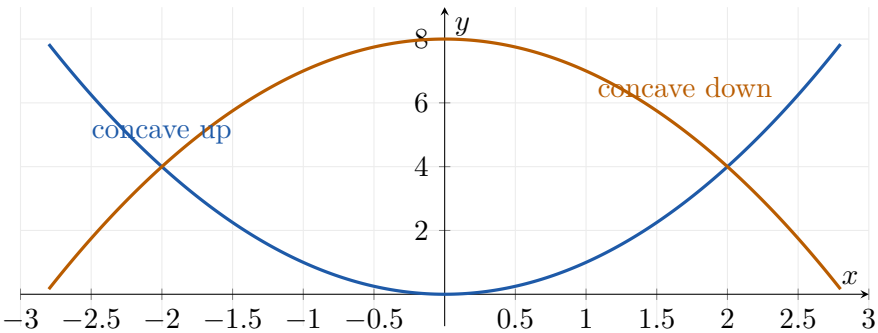
6.4 Second derivative and concavity

The second derivative is the derivative of the derivative:

$$f''(x) = \frac{\mathrm{d}}{\mathrm{d}x} f'(x).$$

It measures how the slope is changing.

- If $f''(x) > 0$, the graph is concave up.
- If $f''(x) < 0$, the graph is concave down.



6.5 Optimization in one variable

Optimization means finding the best value: maximum or minimum.

Simple minimization

Minimize

$$f(x) = x^2 - 6x + 10.$$

Compute the derivative:

$$f'(x) = 2x - 6.$$

Set it equal to zero:

$$2x - 6 = 0 \quad \Rightarrow \quad x = 3.$$

Second derivative:

$$f''(x) = 2 > 0.$$

So $x = 3$ is a minimum. The minimum value is

$$f(3) = 9 - 18 + 10 = 1.$$

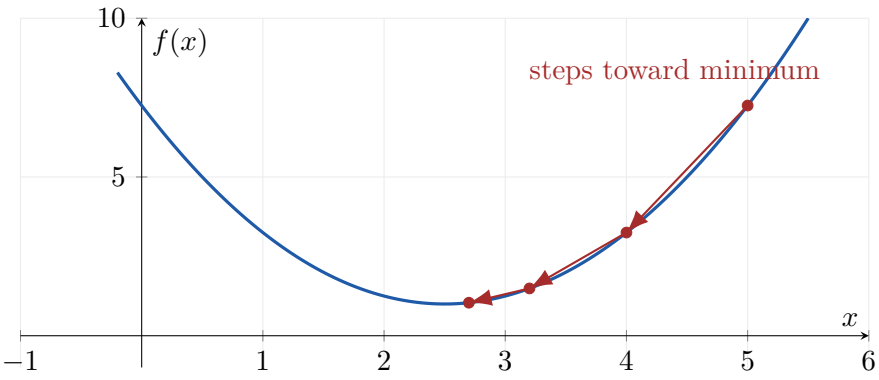
6.6 Connection to gradient descent

In machine learning and optimization, a very simple idea is:

$$\text{new } x = \text{old } x - \eta f'(x),$$

where $\eta > 0$ is a small step size. This is one-dimensional gradient descent.

If $f'(x) > 0$, the function increases to the right, so we move left. If $f'(x) < 0$, the function decreases to the right, so we move right.



Practice

For each function, find critical points and classify them if possible.

1. $f(x) = x^2 + 4x + 1$
2. $g(x) = x^3 - 6x^2 + 9x$
3. $h(x) = x^4 - 4x^2$

7 Linear approximation and Taylor idea

7.1 Why approximate?

Computers often approximate complicated functions by simpler ones. The derivative tells us the best linear approximation near a point.

Linear approximation

If f is differentiable near a, then near a:

$$f(x) \approx f(a) + f'(a)(x - a).$$

This is the tangent line approximation.

Approximate a square root

Approximate $\sqrt{4.1}$.
Use $f(x) = \sqrt{x}$, and choose $a = 4$ because $\sqrt{4} = 2$.

$$f'(x) = \frac{1}{2\sqrt{x}}, \qquad f'(4) = \frac{1}{4}.$$

Therefore

$$\sqrt{x} \approx 2 + \frac{1}{4}(x - 4).$$

For $x = 4.1$:

$$\sqrt{4.1} \approx 2 + \frac{1}{4}(0.1) = 2.025.$$

7.2 Taylor polynomials

The linear approximation uses the first derivative. Taylor polynomials use more derivatives.

Second-order Taylor approximation

Near $x = a$,

$$f(x) \approx f(a) + f'(a)(x - a) + \frac{f''(a)}{2}(x - a)^2.$$

The more terms we use, the better the approximation often becomes near a.

7.3 Important Taylor approximations near zero

For small x:

$$e^x \approx 1 + x + \frac{x^2}{2},$$

$$\sin x \approx x - \frac{x^3}{6},$$

$$\cos x \approx 1 - \frac{x^2}{2},$$

$$\ln(1 + x) \approx x - \frac{x^2}{2}.$$

Common mistake

Taylor approximations are local. A formula that works well near 0 may be bad far from 0.

Practice

Use linear approximation.

1. Approximate $\sqrt{9.2}$ using $a = 9$.
2. Approximate $\ln(1.05)$ using $\ln(1 + x) \approx x$.
3. Approximate $e^{0.03}$ using $e^x \approx 1 + x$.

8 Antiderivatives: undoing derivatives

8.1 The basic idea

If differentiation takes a function and gives its rate of change, then antidifferentiation goes in the opposite direction.

Antiderivative

A function F is an antiderivative of f if

$$F'(x) = f(x).$$

Example:

$$F(x) = x^3$$

is an antiderivative of

$$f(x) = 3x^2,$$

because

$$\frac{\mathrm{d}}{\mathrm{d}x}x^3 = 3x^2.$$

8.2 The constant of integration

If $F'(x) = f(x)$, then

$$(F(x) + C)' = f(x)$$

for any constant C. Therefore indefinite integrals always include C.

Indefinite integral

$$\int f(x)\,\mathrm{d}x = F(x) + C,$$

where $F'(x) = f(x)$.

Basic antiderivative

$$\int 2x\,\mathrm{d}x = x^2 + C.$$

Check:

$$\frac{\mathrm{d}}{\mathrm{d}x}(x^2 + C) = 2x.$$

Common mistake

Do not forget $+C$ in an indefinite integral. Without $+C$, you give only one antiderivative, not all of them.

8.3 Basic integration rules

Power rule for integration

If $n \neq -1$, then

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C.$$

Examples:

$$\int x^2 \, dx = \frac{x^3}{3} + C,$$

$$\int 5x^4 \, dx = x^5 + C,$$

$$\int \sqrt{x} \, dx = \int x^{1/2} \, dx = \frac{x^{3/2}}{3/2} + C = \frac{2}{3}x^{3/2} + C.$$

8.4 Special case: $1/x$

The power rule does not work for $n = -1$, because it would require division by zero.

Instead:

$$\int \frac{1}{x} \, dx = \ln|x| + C.$$

8.5 Common antiderivatives

Function	Antiderivative
$x^n, \ n \neq -1$	$\frac{x^{n+1}}{n+1} + C$
$1/x$	$\ln\|x\| + C$
e^x	$e^x + C$
$\cos x$	$\sin x + C$
$\sin x$	$-\cos x + C$
$\sec^2 x$	$\tan x + C$

Polynomial integral

$$\int (6x^2 - 4x + 7) \, dx = 2x^3 - 2x^2 + 7x + C.$$

Practice

Compute the indefinite integrals.

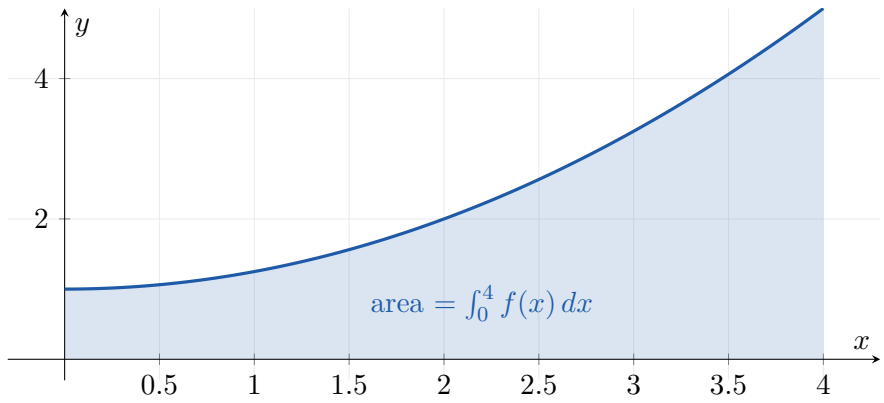
1. $\int 3x^2 \, dx$
2. $\int (4x^3 - 2x + 5) \, dx$
3. $\int \frac{1}{x} \, dx$
4. $\int e^x \, dx$

5. $\displaystyle\int \cos x \, \mathrm{d}x$

9 Definite integrals: area and accumulation

9.1 From rectangles to area

A definite integral measures accumulated quantity over an interval. Geometrically, when $f(x) \geq 0$, it gives the area under the graph.



Definite integral

$$\int_a^b f(x)\,\mathrm{d}x$$

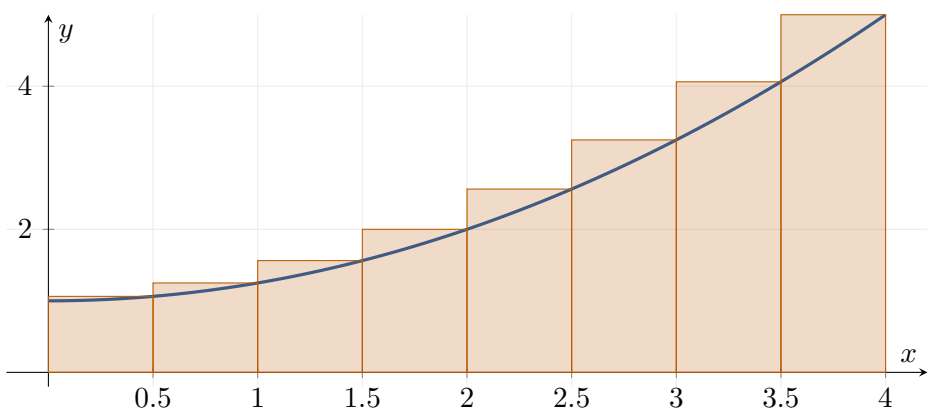
represents the signed accumulated area between the graph of f and the x-axis from $x = a$ to $x = b$.

9.2 Riemann sums

To define area precisely, split the interval $[a, b]$ into small pieces. On each piece, approximate the area by a rectangle.

$$\text{area} \approx \sum_{i=1}^{n} f(x_i^*)\Delta x.$$

As the rectangles become thinner, the approximation approaches the integral.



9.3 Signed area

If $f(x) > 0$, the integral contributes positive area. If $f(x) < 0$, it contributes negative area.

Simple signed area

For $f(x) = x$,

$$\int_{-1}^{1} x \, \mathrm{d}x = 0.$$

The negative area on $[-1, 0]$ cancels the positive area on $[0, 1]$.

9.4 Basic properties of definite integrals

$$\int_a^a f(x) \, \mathrm{d}x = 0,$$

$$\int_a^b f(x) \, \mathrm{d}x = -\int_b^a f(x) \, \mathrm{d}x,$$

$$\int_a^b (f(x) + g(x)) \, \mathrm{d}x = \int_a^b f(x) \, \mathrm{d}x + \int_a^b g(x) \, \mathrm{d}x,$$

$$\int_a^b cf(x) \, \mathrm{d}x = c \int_a^b f(x) \, \mathrm{d}x.$$

Practice

Interpret each integral as signed area.

1. $\displaystyle\int_0^3 2 \, \mathrm{d}x$
2. $\displaystyle\int_{-2}^2 x \, \mathrm{d}x$
3. $\displaystyle\int_0^1 x^2 \, \mathrm{d}x$

10 The Fundamental Theorem of Calculus

10.1 The bridge between derivative and integral

The Fundamental Theorem of Calculus says that derivatives and integrals are inverse operations in a precise sense.

Fundamental Theorem of Calculus, Part 1

If

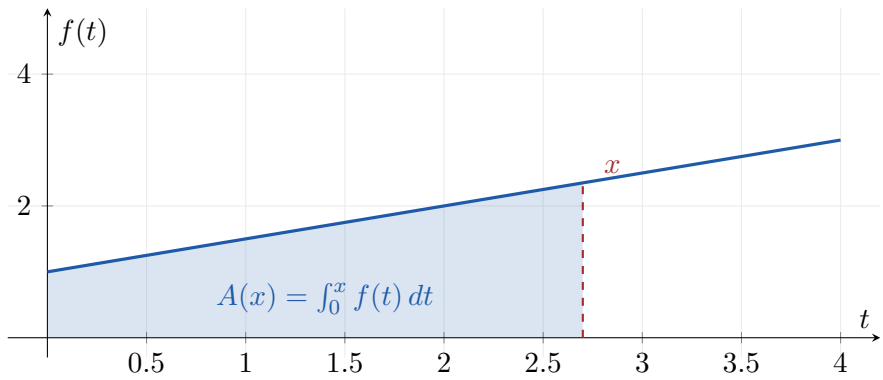
$$A(x) = \int_a^x f(t)\,\mathrm{d}t,$$

then

$$A'(x) = f(x),$$

assuming f is continuous.

This means: if $A(x)$ accumulates area under f, then the rate of change of accumulated area is the current height $f(x)$.



10.2 Computing definite integrals using antiderivatives

Fundamental Theorem of Calculus, Part 2

If $F'(x) = f(x)$, then

$$\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a).$$

Computing a definite integral

Compute

$$\int_0^3 x^2\,\mathrm{d}x.$$

An antiderivative of x^2 is

$$F(x) = \frac{x^3}{3}.$$

Therefore

$$\int_0^3 x^2\,\mathrm{d}x = F(3) - F(0) = \frac{27}{3} - 0 = 9.$$

10.3 Why the theorem matters

Without the theorem, definite integrals require limits of sums. With the theorem, many integrals become simple antiderivative calculations.

Common mistake

The expression

$$\int_a^b f(x)\,\mathrm{d}x$$

is a number. The expression

$$\int f(x)\,\mathrm{d}x$$

is a family of functions. Do not confuse definite and indefinite integrals.

Practice

Compute using antiderivatives.

1. $\displaystyle\int_0^2 3x^2\,\mathrm{d}x$
2. $\displaystyle\int_1^4 \frac{1}{x}\,\mathrm{d}x$
3. $\displaystyle\int_0^\pi \sin x\,\mathrm{d}x$
4. $\displaystyle\int_{-1}^1 (x^2+1)\,\mathrm{d}x$

11 Techniques of integration

11.1 Substitution

Substitution is the reverse of the chain rule.

Substitution rule

If $u = g(x)$, then

$$\int f(g(x))g'(x)\,\mathrm{d}x = \int f(u)\,\mathrm{d}u.$$

Substitution example

Compute

$$\int 2x(x^2+1)^5\,\mathrm{d}x.$$

Let

$$u = x^2 + 1.$$

Then

$$\mathrm{d}u = 2x\,\mathrm{d}x.$$

So

$$\int 2x(x^2+1)^5\,\mathrm{d}x = \int u^5\,\mathrm{d}u = \frac{u^6}{6} + C.$$

Return to x:

$$\boxed{\frac{(x^2+1)^6}{6} + C.}$$

Common mistake

After substitution, every part of the integral must be changed into u-language. Do not leave a mixture of x and u unless you are in the middle of the work and about to fix it.

11.2 Integration by parts

Integration by parts is the reverse of the product rule.

Integration by parts

$$\int u\,\mathrm{d}v = uv - \int v\,\mathrm{d}u.$$

Integration by parts example

Compute

$$\int xe^x\,\mathrm{d}x.$$

Choose

$$u = x, \qquad \mathrm{d}v = e^x\,\mathrm{d}x.$$

Then

$$\mathrm{d}u = \mathrm{d}x, \qquad v = e^x.$$

So

$$\int xe^x \, \mathrm{d}x = xe^x - \int e^x \, \mathrm{d}x = xe^x - e^x + C.$$

11.3 Partial fractions, very briefly

Partial fractions help integrate rational functions.

Partial fractions

Compute

$$\int \frac{1}{x(x+1)} \, \mathrm{d}x.$$

Write

$$\frac{1}{x(x+1)} = \frac{A}{x} + \frac{B}{x+1}.$$

Then

$$1 = A(x+1) + Bx = (A+B)x + A.$$

So

$$A = 1, \qquad A + B = 0 \Rightarrow B = -1.$$

Thus

$$\int \frac{1}{x(x+1)} \, \mathrm{d}x = \int \left(\frac{1}{x} - \frac{1}{x+1} \right) \mathrm{d}x = \ln|x| - \ln|x+1| + C.$$

Practice

Compute.

1. $\displaystyle\int 6x(x^2+4)^2 \, \mathrm{d}x$
2. $\displaystyle\int x \cos x \, \mathrm{d}x$
3. $\displaystyle\int \frac{1}{x(x+2)} \, \mathrm{d}x$

12 Numerical integration for computer scientists

12.1 Why numerical integration?

Sometimes an exact antiderivative is hard or impossible to express using elementary functions. Computers then approximate integrals numerically.

12.2 Left and right rectangle rules

Split $[a, b]$ into n subintervals of width

$$\Delta x = \frac{b-a}{n}.$$

The left rectangle rule is

$$\int_a^b f(x)\,\mathrm{d}x \approx \Delta x \sum_{i=0}^{n-1} f(a + i\Delta x).$$

The right rectangle rule is

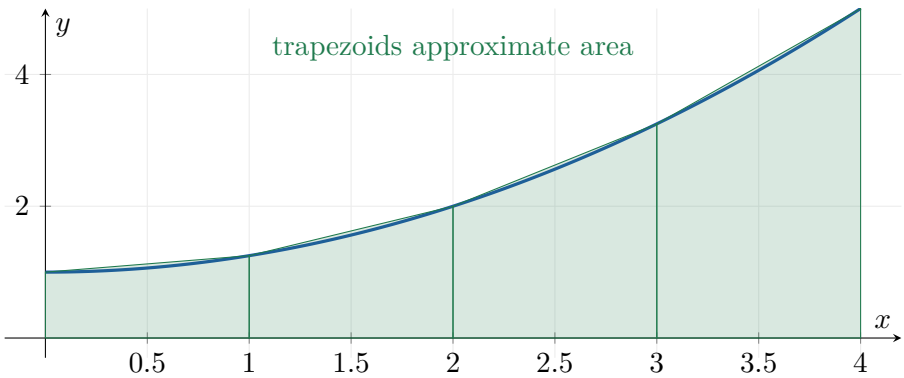
$$\int_a^b f(x)\,\mathrm{d}x \approx \Delta x \sum_{i=1}^{n} f(a + i\Delta x).$$

12.3 Trapezoidal rule

Instead of rectangles, use trapezoids.

Trapezoidal rule

$$\int_a^b f(x)\,\mathrm{d}x \approx \frac{\Delta x}{2}\left(f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(x_n)\right).$$



12.4 A simple algorithmic view

To approximate $\int_a^b f(x)\,\mathrm{d}x$:

1. choose the number of subintervals n;
2. compute $\Delta x = (b-a)/n$;
3. evaluate f at selected grid points;

4. sum the weighted values;
5. multiply by Δx or $\Delta x/2$, depending on the method.

12.5 Error intuition

More subintervals usually give a better approximation, but also require more evaluations of f. This creates a standard computational trade-off: accuracy versus cost.

Practice

Use the trapezoidal rule with $n = 4$ to approximate

$$\int_0^2 x^2 \, \mathrm{d}x.$$

Then compare with the exact value.

13 Multivariable calculus: the minimum needed later

13.1 Functions of several variables

A function can depend on more than one input:

$$f(x, y) = x^2 + y^2.$$

This function takes a pair (x, y) and returns one number.

13.2 Partial derivatives

A partial derivative measures change with respect to one variable while the others are treated as constants.

Partial derivative

For $f(x, y)$,

$$\frac{\partial f}{\partial x}$$

means: differentiate with respect to x, treating y as constant.

$$\frac{\partial f}{\partial y}$$

means: differentiate with respect to y, treating x as constant.

Partial derivatives

Let

$$f(x, y) = x^2 + 3xy + y^2.$$

Then

$$\frac{\partial f}{\partial x} = 2x + 3y,$$

because y is treated as constant.
Also

$$\frac{\partial f}{\partial y} = 3x + 2y.$$

13.3 Gradient

The gradient collects all first partial derivatives into a vector.

Gradient

For $f(x, y)$,

$$\nabla f(x, y) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right).$$

For

$$f(x, y) = x^2 + y^2,$$

we have

$$\nabla f(x, y) = (2x, 2y).$$

The gradient points in the direction of fastest increase. Negative gradient points in the direction of fastest decrease.

13.4 Gradient descent in several variables

Gradient descent becomes

$$\vec{x}_{new} = \vec{x}_{old} - \eta \nabla f(\vec{x}_{old}).$$

This formula is central in many optimization and machine learning methods.

13.5 Hessian

The Hessian matrix collects second partial derivatives:

$$H_f(x, y) = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix}.$$

It describes curvature.

Practice

For each function, compute the gradient.

1. $f(x, y) = x^2 + 4y^2$
2. $g(x, y) = xy + x$
3. $h(x, y) = e^x + y^3$

14 Common beginner mistakes and how to avoid them

14.1 Mistake 1: treating notation as decoration

Symbols carry meaning. In

$$\int_a^b f(x)\,\mathrm{d}x,$$

all parts matter:

- a and b are the limits of integration;
- $f(x)$ is the function being accumulated;
- $\mathrm{d}x$ tells us the variable of integration.

14.2 Mistake 2: substituting too early in limits

If substitution gives $0/0$, do not stop. Simplify first.

14.3 Mistake 3: forgetting the chain rule

For

$$(5x-1)^7,$$

the derivative is not only

$$7(5x-1)^6.$$

It is

$$35(5x-1)^6.$$

14.4 Mistake 4: forgetting $+C$

$$\int 2x\,\mathrm{d}x = x^2 + C,$$

not just x^2.

14.5 Mistake 5: confusing area with signed area

If a function goes below the x-axis, the definite integral counts that part negatively.

14.6 Mistake 6: ignoring domain

Expressions such as $\ln x$, $\sqrt{x}$, and $1/x$ have restrictions. Always check whether the expression is defined.

15 A compact formula sheet

Limits

$$\lim_{x\to a} f(x) = L \quad \text{means} \quad f(x) \text{ approaches } L \text{ as } x \text{ approaches } a.$$

For continuous functions:

$$\lim_{x\to a} f(x) = f(a).$$

Continuity

$$f \text{ continuous at } a \iff f(a) \text{ exists, } \lim_{x\to a} f(x) \text{ exists, and } \lim_{x\to a} f(x) = f(a).$$

Derivatives

$$f'(a) = \lim_{h\to 0} \frac{f(a+h) - f(a)}{h}.$$

$$\frac{\mathrm{d}}{\mathrm{d}x} x^n = nx^{n-1}.$$

$$(f+g)' = f' + g', \qquad (cf)' = cf'.$$

$$(fg)' = f'g + fg'.$$

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}.$$

$$(f(g(x)))' = f'(g(x))g'(x).$$

Integrals

$$\int x^n \,\mathrm{d}x = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1.$$

$$\int \frac{1}{x} \,\mathrm{d}x = \ln|x| + C.$$

$$\int_a^b f(x) \,\mathrm{d}x = F(b) - F(a), \quad F' = f.$$

Optimization

Critical points occur where

$$f'(x) = 0$$

or where $f'(x)$ does not exist.

Multivariable basics

$$\nabla f = \left(\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n}\right).$$

$$\vec{x}_{new} = \vec{x}_{old} - \eta \nabla f(\vec{x}_{old}).$$

16 Selected solutions to practice problems

Slope practice

1. $m = (5-1)/(2-0) = 2$.
2. $m = (-5-3)/(3-(-1)) = -8/4 = -2$.
3. $m = (10-7)/(2-2) = 3/0$, undefined. The line is vertical.

Limit practice

1. $\lim_{x\to 4}(3x-7) = 12-7 = 5$.
2. $\lim_{x\to -1}(x^2+2x+5) = 1-2+5 = 4$.
3. $\frac{x^2-9}{x-3} = x+3$ for $x \neq 3$, so the limit is 6.
4. $\lim_{x\to 0^+} 1/x^2 = +\infty$.
5. $\lim_{x\to\infty} 5/x = 0$.

Derivative practice

1. $\frac{\mathrm{d}}{\mathrm{d}x}(5x^4-2x+9) = 20x^3-2$.
2. $\frac{\mathrm{d}}{\mathrm{d}x}(x^2+1)^3 = 3(x^2+1)^2 \cdot 2x = 6x(x^2+1)^2$.
3. $\frac{\mathrm{d}}{\mathrm{d}x}(x^2e^x) = 2xe^x + x^2e^x = e^x(x^2+2x)$.
4. $\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\ln x}{x}\right) = \frac{(1/x)x - \ln x}{x^2} = \frac{1-\ln x}{x^2}$.
5. $\frac{\mathrm{d}}{\mathrm{d}x}\cos(4x) = -\sin(4x)\cdot 4 = -4\sin(4x)$.

Integral practice

1. $\int 3x^2\,\mathrm{d}x = x^3 + C$.
2. $\int(4x^3-2x+5)\,\mathrm{d}x = x^4 - x^2 + 5x + C$.
3. $\int \frac{1}{x}\,\mathrm{d}x = \ln|x| + C$.
4. $\int e^x\,\mathrm{d}x = e^x + C$.
5. $\int \cos x\,\mathrm{d}x = \sin x + C$.

Definite integral practice

1. $\int_0^2 3x^2\,\mathrm{d}x = [x^3]_0^2 = 8$.
2. $\int_1^4 \frac{1}{x}\,\mathrm{d}x = [\ln x]_1^4 = \ln 4$.
3. $\int_0^\pi \sin x\,\mathrm{d}x = [-\cos x]_0^\pi = 2$.
4. $\int_{-1}^1 (x^2+1)\,\mathrm{d}x = [x^3/3 + x]_{-1}^1 = 8/3$.